Castle at the Crossroads: A Machine Learning Approach to Generic Mixture in the Nineteenth-Century Gothic Novel

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Introduction

By the late twentieth century, genre, in literary study, had transformed from a descriptive taxonomy of different kinds of writing to an evolving mixture of tropes and conventions whose boundaries shift and reform across time (eg Cohen 1986, Derrida 1980). Computational approaches to literature, however, still frequently lean towards treating genre as a strong classificatory system whose boundaries can be denoted within the rows of a spreadsheet (Allison et. al. 2011, Jockers 2013, Schöch 2017, Sobchuk and Šeļa 2024). Although partly due to the nature of the data structures in which we store information related to individual books, this approach to genre perhaps owes more to the wealth of classification methods available to computational literary models: given a classificatory system to arrange literature, our impulse as digital humanists has been to use computational methods to either replicate our system of genre classification, or to introduce a different, yet equally taxonomic system based solely on endogamous textual features.

Our project draws on contemporary literary theories of genre in order to model individual novels as heterogeneous mixtures of styles and conventions from pre-existing generic modes. Building on a recent trend in the field to use machine learning to complicate our picture of genre (Elliot 2017, Underwood 2018, Piper 2018, Tello 2021, King 2021, Algee-Hewitt et. al. 2023), we introduce a new approach to modeling genre, one that takes seriously the literary theoretical understanding of genre as a set of contingent categories whose mixture and remixture over time creates contingent similarities rather than discrete categories. The nineteenth-century English Gothic novel is an ideal case study for this project. Not only is the emergence of the genre in 1764 well understood, its subsequent transformations over the nineteenth century and its hybridity with other kinds of writing, such as sensation fiction, detective novels, thrillers, and satires has been long documented. Our project asks how, given the understanding of the Gothic as a mixture of pre-existing generic writing, we can use a computationally-based modeling approach to decompose this mixture and resurrect the specific generic investments of a single novel.

Corpus

Drawing on our expert knowledge of nineteenth-century Gothic writing and its adjacent genres, we selected nine different novelistic genres whose styles and conventions are mixed into the Gothic novel across the nineteenth century. We then gathered samples of these genres from the Chadwyck Healey corpus of nineteenth-century Literature and the British Library corpus of nineteenth-century fiction. We selected our samples based on the literary scholarship on each of these genres, adding novels listed by primary critics of each. We collected novels of nine different novelistic genres apart from the Gothic: 35 courtship novels (Green 1991, Yeazell 1991); 57 distant-historical novels, which we define as novels taking place in the ancient and medieval periods; 94 recent-historical novels, which we define as taking place in the late Early Modern through eighteenth century periods (Lukács 1955); 46 novels of manners (Michie 2011, Macheski 2012); 28 national tales (Trumpener 1997); 38 realist novels (Watt 1957, Barthes 1968, Woloch 2003, Greiner 2013); 53 works of sea fiction (Cohen 2010); 45 works of sensation fiction (Pykett 1994); and 94 silver fork novels (Hughes 1992, Wagner 2009, Copeland 2012). For our comparative Gothic corpus, we selected 35 Gothic novels, published between 1764 (The Castle of Otranto) and 1897 (Dracula). The Gothic novel offers the ideal test case for our theory, as the authors associated with the genre theorized its development themselves as a mixture of heterogeneous genres (Garber 1974).

Method

The key insight of our method is that we train a model to recognize genre in the Gothic novel wi-

thout including Gothic texts as part of our training data. Instead, we classify sections of our corpus of Gothic novels using a classifier trained on our corpora of contemporaneous genres. We divide the texts in our comparative corpus into passages of about 500 words (as we respected sentence boundaries in dividing our texts, we selected the number of complete sentences the most closely approximated 500 words). We then fine-tune a BERT based classification model to differentiate between the genres in our comparison corpus. A withheld test set of passages reveals that our model is working reasonable well at differentiating our comparative genres (Figure 1).

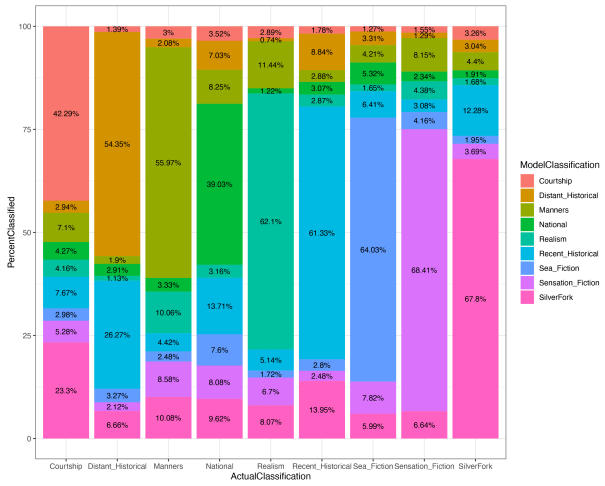

Figure 1: Confusion plot of BERT model classification of our comparative genres. Overall success rate is 60% compared to chance at 11%. Although this seems low for some classification tasks, our overall approach to genre relies on the fact that even these comparative genres are themselves mixtures of multiple types, hence the difficulties in classifying these genres reflect ambiguities in the underlying data rather than any flaws in our model. For the purposes of our experiment, it is important to note that the majority of passages in each genre classifies into the correct category.

We then use this model to classify passages of the same length in our Gothic corpus. As classifying any given passage as "Gothic" is not an option given the training of the model, our method instead returns individual passages classified into one of our comparative genre categories. We can then aggregate the passages for each novel to calculate the specific mixture that defines it. Using this method, we can not only see the makeup of different novels in our corpus, but we can identify historical patterns in the similarities of the mixtures. Does the Gothic novel of the 1790s rely on a particular combination of genre "borrowings" and does this change as the Gothic evolves over time? Given the transformation of the Gothic as a genre in and of itself, as the set of recognizable tropes that define it change periodically from the late eighteenth to the twentieth century (Sedgwick 1980, Groom 2012), our project is particularly adept at uncovering the minutia of these changes.

Results

Visualizing our mixture results as a series of aligned radial plots of individual novels allows us to graphically display the similarities and differences of the mixtures that characterize novels in our Gothic corpus. For example, the plot of Dracula (figure 2), reveals its strong investment in realism, recent historical fiction, novels of manners, and sensation fiction (with an important component of sea fiction for the passages describing the doomed voyage of the Demeter).

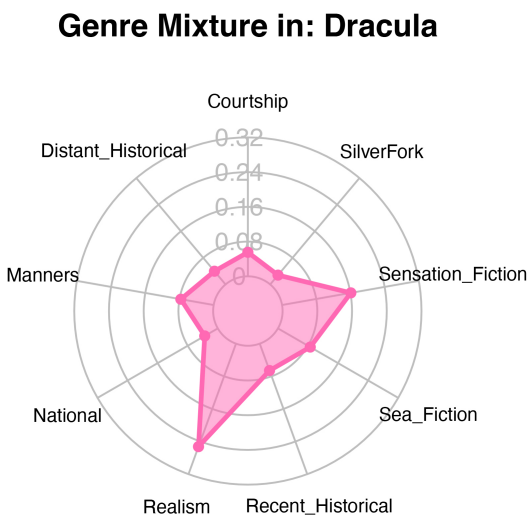

Figure 2: Plot of generic mixture in Dracula by Bram Stoker. The individual radial areas indicated the percentage of the novel classified into each comparative genre by our model.

Contrast this late nineteenth-century novel with the early work of Radcliffe from the 1790s (figure 3).

Figure 3: Radial plots showing the generic mixture of three Ann Radcliffe novels: The Mysteries of Udolpho (1794); The Italian (1797) and The Romance of the Forest (1791).

Radcliffe's novels evidence a strong investment in the recent historical novel (this is, perhaps, unsurprising, given that many of her novels are set one or two hundred years before Radcliffe's own time). Also notable is the strong presence of the Silver Fork novel in each, a trend that correlates with Radcliffe's focus on wealthy, sometimes aristocratic, property-inheriting families and their respective Gothic heroines.

Overall, the results of our modeling experiment reveal both individual facets of the novels in our corpus that align well with their critical reception, as well as strong overall trends in the shifting nature of the Gothic over the century. Our project dives deeply into individual novels, linking passages with specific classifications to their contents, and, at the same time, explores these trends at the scale of the century, including an overall fall of investment in the recent historical novel (post Radcliffe), a mid-century bump in the inclusion of distant historical fiction into the Gothic, and a surprising increase in the number of passages classified as realism. The last, we hypothesize, is related to the transformation of the Gothic from a standalone genre, to itself a series of tropes that can be included into other kinds of writing in order to establish the mood of a particular scene.

Our project offers a new way of understanding genre computationally, not as a set of classifications into which writing can be neatly slotted, but instead an evolving set of relationships between kinds of writing, in which subjects and styles blend into new forms at the scale of the passage. This not only offers important insight into a genre such as the Gothic, providing an accounting of the specific investments of the novels in other genre categories, but also offers a reinterpretation of genre history, particularly within the digital humanities.

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